

Nearest-Neighbor Escape Probability of SRW on $\mathbb{Z}^2$

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Problem

Consider a simple random walk on $\mathbb{Z}^2$ starting at the origin $(0,0)$. What is the probability that it hits $(1,0)$ before it returns to $(0,0)$?

In probability notation, we want

$$\mathbb{P}(\tau_{(1,0)} < \tau_{(0,0)}^+ \mid X_0 = (0,0)),$$

where

$$\tau_{(1,0)} = \inf\{t \geq 0 : X_t = (1,0)\}, \quad \tau_{(0,0)}^+ = \inf\{t > 0 : X_t = (0,0)\}.$$

Equivalently, this is the escape probability of the random walk from $(0,0)$ to $(1,0)$ before returning.

Electric Network Interpretation

Consider the infinite network $\mathbb{Z}^2$ with unit resistance on each edge. If we can compute the effective resistance between the nodes $(0,0)$ and $(1,0)$, then the escape probability is given by

$$\mathbb{P}(\tau_{(1,0)} < \tau_{(0,0)}^+) = \frac{1}{4R_{\text{eff}}((0,0), (1,0))}.$$

So the whole problem reduces to proving

$$R_{\text{eff}}((0,0), (1,0)) = \frac{1}{2}.$$

Two-Flow Construction on the Infinite Network

Let

$$o = (0,0), \quad a = (1,0).$$

At first sight, one wants to consider two networks:

- **Network 1:** Put 1A of current into o and take 1A of current out from “infinitely far away”.

- **Network 2:** Put 1A of current into the “infinitely far away” and take 1A of current out from a .

If one were allowed to work with these two infinite networks directly, then the answer would look immediate.

For the first network, by symmetry at o , the current splits equally into the four adjacent edges, so the current through the edge

$$o \rightarrow a$$

should be

$$I_{o \rightarrow a}^{(1)} = \frac{1}{4}.$$

For the second network, by symmetry at a , the current should also split equally among the four edges adjacent to a , so the current through the same edge should again be

$$I_{o \rightarrow a}^{(2)} = \frac{1}{4}.$$

Now if one adds the two networks, one gets a new network with 1A entering at o and 1A leaving at a . By linearity of electric networks, the current through the edge $o \rightarrow a$ should then be

$$I_{o \rightarrow a} = I_{o \rightarrow a}^{(1)} + I_{o \rightarrow a}^{(2)} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

Since the edge resistance is 1, this would give

$$V(o) - V(a) = 1 \cdot I_{o \rightarrow a} = \frac{1}{2},$$

hence

$$R_{\text{eff}}(o, a) = \frac{1}{2}, \quad \mathbb{P}(\tau_a < \tau_o^+) = \frac{1}{4R_{\text{eff}}(o, a)} = \frac{1}{2}.$$

This argument is very natural, but there is a real problem here: the phrase “infinitely far away” has not been defined, and in a recurrent graph like $\mathbb{Z}^2$ one cannot simply superpose two infinite networks without justification.

So the real question is how to make this idea rigorous.

Finite-Box Approximation

Let

$$B_n^{(0)} = [-n, n]^2 \cap \mathbb{Z}^2.$$

Wire all points of $\partial B_n^{(0)}$ into a single boundary vertex.

We first define two finite networks on this wired box.

- **Network 1 on $B_n^{(0)}$:** put 1A into o and take 1A out from the wired boundary.
- **Network 2 on $B_n^{(0)}$:** put 1A into the wired boundary and take 1A out from a .

Let

$i_n^{(1)}$ be the current of Network 1 on $B_n^{(0)}$,

and

$i_n^{(2)}$ be the current of Network 2 on $B_n^{(0)}$.

Their sum

$$\widetilde{i}_n := i_n^{(1)} + i_n^{(2)}$$

is exactly the unit current flow from o to a on the wired box $B_n^{(0)}$.

So the philosophy is exactly the same as before. The only issue is now to justify what happens as $n \rightarrow \infty$.

Network 1

For Network 1, the box $B_n^{(0)}$ is centered at o , so the four edges adjacent to o are symmetric. Therefore the current leaving o splits equally among these four edges. In particular,

$$i_n^{(1)}(o, a) = \frac{1}{4} \quad \text{for every } n.$$

Network 2

The second network is harder because the box $B_n^{(0)}$ is not centered at a . So even though in the infinite picture one expects symmetry around a , at finite n there is no exact symmetry forcing

$$i_n^{(2)}(o, a) = \frac{1}{4}.$$

This is the point where my original argument was incomplete.

A wrong first attempt is to consider

$$v_n(x) := \mathbb{P}_x(\tau_{\partial B_n^{(0)}} < \tau_a),$$

which is the unit voltage function with boundary value 1 at $\partial B_n^{(0)}$ and 0 at a . But since simple random walk on $\mathbb{Z}^2$ is recurrent, for each fixed x we have

$$v_n(x) \rightarrow 0 \quad (n \rightarrow \infty),$$

so this normalization is useless for the present problem.

The correct normalization is to multiply by the effective resistance from a to the boundary.

Normalized Voltage for Network 2

Let

$$r_n := R_{\text{eff}}(a \leftrightarrow \partial B_n^{(0)}).$$

Define

$$V_n^{(0)}(x) := r_n v_n(x).$$

Then $V_n^{(0)}$ is the voltage function for Network 2 on the box $B_n^{(0)}$, normalized by

$$V_n^{(0)}(a) = 0.$$

Equivalently, $V_n^{(0)}$ satisfies

$$\begin{aligned} V_n^{(0)}(a) &= 0, \\ \Delta V_n^{(0)}(x) &= 0 \quad \text{for } x \in B_n^{(0)} \setminus \{a\}, \end{aligned}$$

and the total current extracted at a is 1, namely

$$\sum_{y \sim a} (V_n^{(0)}(y) - V_n^{(0)}(a)) = 1.$$

Since $V_n^{(0)}(a) = 0$, this becomes

$$\sum_{y \sim a} V_n^{(0)}(y) = 1.$$

If one could prove that, for each fixed x ,

$$V_n^{(0)}(x) \longrightarrow V(x),$$

and that this limit agrees near a with the symmetric limit coming from a box centered at a , then necessarily the four neighbors of a would have the same limit and sum to 1, hence each limit would be $1/4$. That would give exactly what we want.

A Symmetric Comparison Network

Define the box centered at a by

$$B_n^{(a)} := ([-n+1, n+1] \times [-n, n]) \cap \mathbb{Z}^2.$$

Again wire $\partial B_n^{(a)}$ into a single boundary vertex.

Define $V_n^{(a)}$ to be the voltage for the unit current flow from the wired boundary of $B_n^{(a)}$ to a , normalized by

$$V_n^{(a)}(a) = 0.$$

So $V_n^{(a)}$ satisfies

$$\begin{aligned} V_n^{(a)}(a) &= 0, \\ \Delta V_n^{(a)}(x) &= 0 \quad \text{for } x \in B_n^{(a)} \setminus \{a\}, \end{aligned}$$

and

$$\sum_{y \sim a} (V_n^{(a)}(y) - V_n^{(a)}(a)) = 1.$$

Since the box $B_n^{(a)}$ is centered at a , the four neighbors of a are symmetric, hence

$$V_n^{(a)}(0, 0) = V_n^{(a)}(2, 0) = V_n^{(a)}(1, 1) = V_n^{(a)}(1, -1) = \frac{1}{4}.$$

In particular,

$$i_{n, \text{sym}}^{(2)}(o, a) = V_n^{(a)}(o) - V_n^{(a)}(a) = \frac{1}{4}.$$

So the remaining problem is:

Show that, for fixed x near a ,

$$V_n^{(0)}(x) - V_n^{(a)}(x) \rightarrow 0?$$

If yes, then

$$V_n^{(0)}(o) \rightarrow \frac{1}{4},$$

therefore

$$i_n^{(2)}(o, a) = V_n^{(0)}(o) - V_n^{(0)}(a) \rightarrow \frac{1}{4}.$$

Green Function Representation

For a finite set $D \subset \mathbb{Z}^2$, let

$$\tau_{\partial D} := \inf\{k \geq 0 : X_k \notin D^\circ\},$$

and define the killed Green function

$$g_D(x, y) := \mathbb{E}_x \left[\sum_{k=0}^{\tau_{\partial D}-1} \mathbf{1}_{\{X_k=y\}} \right].$$

For the unit-resistance network, the voltage of the unit current flow from a to ∂D is

$$U_D(x) = \frac{1}{4} g_D(x, a).$$

Therefore the voltage of the reversed unit current flow from ∂D to a , normalized by $V_D(a) = 0$, is

$$V_D(x) := U_D(a) - U_D(x) = \frac{1}{4} (g_D(a, a) - g_D(x, a)).$$

Hence

$$V_n^{(0)}(x) = \frac{1}{4} (g_{B_n^{(0)}}(a, a) - g_{B_n^{(0)}}(x, a)),$$

and

$$V_n^{(a)}(x) = \frac{1}{4} (g_{B_n^{(a)}}(a, a) - g_{B_n^{(a)}}(x, a)).$$

So the comparison between the two voltage functions is exactly a comparison between two killed Green functions.

Potential Kernel Formula

$$a(z) = \frac{2}{\pi} \log |z| + c_0 + O(|z|^{-2}) \quad (|z| \rightarrow \infty).$$

Standard representation of the killed Green function:

$$g_D(x, a) = \mathbb{E}_x[\mathfrak{a}(X_{\tau_{\partial D}} - a)] - \mathfrak{a}(x - a).$$

Substituting this into the formula for V_D , we obtain

$$V_D(x) = \frac{1}{4} \left(\mathbb{E}_a[\mathfrak{a}(X_{\tau_{\partial D}} - a)] - \mathbb{E}_x[\mathfrak{a}(X_{\tau_{\partial D}} - a)] + \mathfrak{a}(x - a) \right).$$

Apply this formula to the two domains $D = B_n^{(0)}$ and $D = B_n^{(a)}$.

Comparison of the two boxes

Let

$$D_0 := B_n^{(0)}, \quad D_a := B_n^{(a)}.$$

Recall that

$$V_D(x) = \frac{1}{4} \left(\mathbb{E}_a[\mathfrak{a}(X_{\tau_D} - a)] - \mathbb{E}_x[\mathfrak{a}(X_{\tau_D} - a)] + \mathfrak{a}(x - a) \right).$$

Therefore

$$\begin{aligned} V_{D_0}(x) - V_{D_a}(x) &= \frac{1}{4} \left(\mathbb{E}_a[\mathfrak{a}(X_{\tau_{D_0}} - a)] - \mathbb{E}_a[\mathfrak{a}(X_{\tau_{D_a}} - a)] \right. \\ &\quad \left. - \mathbb{E}_x[\mathfrak{a}(X_{\tau_{D_0}} - a)] + \mathbb{E}_x[\mathfrak{a}(X_{\tau_{D_a}} - a)] \right). \end{aligned}$$

$\mathfrak{a}(x - a)$ has disappeared, since it does not depend on the domain.

For a domain D , write the exit distribution as

$$H_D(y, \xi) := \mathbb{P}_y(X_{\tau_D} = \xi), \quad \xi \in \partial D.$$

Then

$$\mathbb{E}_y[\mathfrak{a}(X_{\tau_D} - a)] = \sum_{\xi \in \partial D} H_D(y, \xi) \mathfrak{a}(\xi - a).$$

Hence

$$\begin{aligned} V_{D_0}(x) - V_{D_a}(x) &= \frac{1}{4} \left[\sum_{\xi \in \partial D_0} H_{D_0}(a, \xi) \mathfrak{a}(\xi - a) - \sum_{\eta \in \partial D_a} H_{D_a}(a, \eta) \mathfrak{a}(\eta - a) \right. \\ &\quad \left. - \sum_{\xi \in \partial D_0} H_{D_0}(x, \xi) \mathfrak{a}(\xi - a) + \sum_{\eta \in \partial D_a} H_{D_a}(x, \eta) \mathfrak{a}(\eta - a) \right]. \end{aligned}$$

Now D_0 and D_a differ only by a shift of the boundary by one lattice unit. Thus the relevant boundary points can be paired as

$$\eta = \xi + u, \quad |u| \leq 1.$$

Moreover, all these boundary points satisfy

$$|\xi - a| \asymp n, \quad |\eta - a| \asymp n.$$

Using the potential kernel asymptotic

$$\mathfrak{a}(z) = \frac{2}{\pi} \log |z| + c_0 + O(|z|^{-2}),$$

we get, uniformly for $|z| \asymp n$ and $|u| \leq 1$,

$$\mathfrak{a}(z + u) - \mathfrak{a}(z) = O\left(\frac{1}{n}\right).$$

$$\mathbb{E}_y[\mathfrak{a}(X_{\tau_{D_0}} - a)] - \mathbb{E}_y[\mathfrak{a}(X_{\tau_{D_a}} - a)] = O\left(\frac{1}{n}\right),$$

for every fixed starting point y in a bounded neighborhood of a . Applying this with $y = a$ and $y = x$, we obtain

$$V_{D_0}(x) - V_{D_a}(x) = O\left(\frac{1}{n}\right).$$

In particular, taking $x = o$,

$$V_n^{(0)}(o) - V_n^{(a)}(o) = O\left(\frac{1}{n}\right) \rightarrow 0.$$

But by the local current computation,

$$V_n^{(a)}(o) = \frac{1}{4}.$$

Hence

$$V_n^{(0)}(o) \rightarrow \frac{1}{4}.$$

Since $V_n^{(0)}(a) = 0$, it follows that

$$i_n^{(2)}(o, a) = V_n^{(0)}(o) - V_n^{(0)}(a) \rightarrow \frac{1}{4}.$$

Passing to the Limit

Now we know:

- for every n ,

$$i_n^{(1)}(o, a) = \frac{1}{4};$$

- and

$$i_n^{(2)}(o, a) \rightarrow \frac{1}{4}.$$

Since

$$\tilde{i}_n = i_n^{(1)} + i_n^{(2)}$$

is the unit current flow from o to a on the wired box $B_n^{(0)}$, the current through the edge $o \rightarrow a$ satisfies

$$\tilde{i}_n(o, a) = i_n^{(1)}(o, a) + i_n^{(2)}(o, a) \rightarrow \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

Because o and a are adjacent and the edge resistance is 1, the voltage drop across that edge is

$$V_n(o) - V_n(a) = \tilde{i}_n(o, a),$$

hence

$$R_{\text{eff}}^{B_n^{(0)}}(o, a) \rightarrow \frac{1}{2}.$$

By wired exhaustion,

$$R_{\text{eff}}^{\mathbb{Z}^2}(o, a) = \lim_{n \rightarrow \infty} R_{\text{eff}}^{B_n^{(0)}}(o, a) = \frac{1}{2}.$$

Therefore

$$\mathbb{P}(\tau_a < \tau_o^+) = \frac{1}{4R_{\text{eff}}^{\mathbb{Z}^2}(o, a)} = \frac{1}{4 \cdot (1/2)} = \frac{1}{2}.$$